# SIH26162

# Persistent Industrial Thermal-Source Detection

Model, experimental progression, final results, and limitations

## Current status

The project converts repeated NASA FIRMS thermal detections into source-level behavioral evidence, then ranks sources that resemble persistent industrial heat. The selected foreign-country model is the NB12b guarded computer-vision plus tabular ensemble.

| Branch | Macro F1 | Macro PR-AUC | Macro ROC-AUC | Worst PR-AUC |
|---|---|---|---|---|
| Compact tabular | 0.7669 | 0.9054 | 0.9199 | 0.8250 |
| Guarded CV plus tabular | 0.8166 | 0.9293 | 0.9481 | 0.8400 |
| All 77 tabular, diagnostic | 0.7347 | 0.8935 | 0.9225 | 0.8489 |

The guarded branch gained 0.0239 macro PR-AUC and 0.0497 macro F1 over the protected compact baseline. It improved PR-AUC in five of six countries and passed all 12 fixed acceptance checks.

The 95% grouped bootstrap interval for the PR-AUC gain was -0.0006 to 0.0515. This passes the non-inferiority guard but crosses zero, so the result does not prove a conventionally significant positive gain.

### Important India status

A historical India evaluation was completed with the superseded NB8 model. The final NB12b guarded model has not been evaluated on India. India was absent from NB12 and NB12b, but it is no longer a pristine project-level holdout.

Prepared from the committed run artifacts on 4 September 2026

# 1. Data and problem framing

NASA FIRMS records satellite thermal anomalies, not confirmed facilities. The audit contains 19,342,072 detections across seven countries from 2019 to 2024. No nulls, malformed dates, or exact duplicate rows were found.

| Country | Detections | Coverage | NOAA-20 |
|---|---|---|---|
| India | 7,594,269 | 2019-2024 | available |
| Angola | 4,869,716 | 2022-2024 | missing |
| Nigeria | 3,914,564 | 2021-2024 | available |
| Iraq | 1,289,983 | 2021-2024 | available |
| Algeria | 744,252 | 2021-2024 | available |
| Indonesia | 501,771 | 2022-2024 | missing |
| Libya | 427,517 | 2021-2024 | available |

## Positive-unlabelled supervision

EOG provides known gas-flare locations. An unmatched source is not a verified negative because it may be a kiln, cement plant, steelworks, missed flare, crop burning, or wildfire. Reported F1 and precision measure agreement with incomplete proxy labels, not verified accuracy for every industrial category.

## Source construction

Detections are grouped into 1 km source cells. Validation uses a separate 10 km block identifier so nearby fragments of one physical site do not cross ordinary fold boundaries. Libya falls from 427,517 detections to 4,824 sources, about 89 detections per source.

## Leakage controls

- Exclude latitude, longitude, country, NASA type, EOG distance, and source identifiers from predictors.
- Exclude NOAA-20 features because Angola and Indonesia lack that instrument.
- Fit preprocessing, thresholds, fusion weights, and branch choices only on training-country folds.
- Use MODIS and VIIRS S-NPP as the uniformly available thermal instruments.

## 2. Tabular model progression

The original Libya plus Algeria study established a strong regional baseline: LightGBM F1 0.691, PR-AUC 0.740, and ROC-AUC 0.930. Those numbers are historical and are not comparable with later six-country protocols.

On the six-country common-window population, the 49-feature LightGBM reached F1 0.523, PR-AUC 0.575, and ROC-AUC 0.974. Intensity was essential, while removing seasonality slightly improved F1. This exposed regional and coverage shortcuts.

### Corrected leave-one-country-out evaluation

| Held-out country | F1 | PR-AUC | Recall |
|---|---|---|---|
| Algeria | 0.596 | 0.660 | 0.475 |
| Angola | 0.165 | 0.231 | 0.281 |
| Indonesia | 0.144 | 0.098 | 0.117 |
| Iraq | 0.641 | 0.568 | 0.638 |
| Libya | 0.667 | 0.757 | 0.550 |
| Nigeria | 0.028 | 0.194 | 0.014 |

Macro F1 was 0.374 and macro PR-AUC was 0.418. Nigeria's collapse showed that geographic transfer, not model size, was the main limitation.

### Robustness changes

- Stage 05c tested country and physical-site weighting. Unweighted training remained strongest.
- Stage 05d nested model selection reached macro F1 0.435 and macro PR-AUC 0.420.
- NB8 regularized domain variants reached macro F1 0.441 and macro PR-AUC 0.460.
- No tabular-only change removed the large country-to-country performance spread.

### Interpretation

Threshold policy improved operational F1, but ranking quality moved less. A more complex recurrent model could not be justified until the country-shift problem and the incomplete-label problem were addressed directly.

# 3. Imagery, temporal modeling, and fusion

A bounded Sentinel-2 L2A and ESA WorldCover workflow created source-centered context chips. NB6 v2 recorded 295 successful chips, four failed chips, and 301 pending candidates. After quality checks and alignment, 294 sources entered the fixed model-comparison panel.

## NB9 multimodal comparison

| Branch | Macro F1 | Macro PR-AUC | Macro ROC-AUC |
|---|---|---|---|
| Early fusion | 0.7916 | 0.9059 | 0.9202 |
| Late fusion | 0.7772 | 0.8710 | 0.9076 |
| Thermal only | 0.6620 | 0.8138 | 0.8491 |
| Image only | 0.6744 | 0.7423 | 0.8604 |

Image-only performance was too weak for an independent CV branch. Imagery became useful only when combined with compact thermal and persistence evidence.

## NB11 temporal and TCN comparison

| Branch | Macro F1 | Macro PR-AUC | Macro ROC-AUC |
|---|---|---|---|
| Nested champion | 0.8140 | 0.9089 | 0.9250 |
| NB9 baseline | 0.7916 | 0.9059 | 0.9202 |
| Temporal descriptors | 0.7933 | 0.9023 | 0.9189 |
| TCN only | 0.7083 | 0.8034 | 0.8570 |

The TCN did not improve ranking enough to justify deployment. Engineered temporal summaries remained useful, but the neural sequence branch added complexity without solving geographic transfer.

## NB12 design

NB12 added frozen CNN embeddings, multispectral summaries, spatial morphology, and a low-capacity visual positive-unlabelled probe. The visual score was allowed to modify the compact tabular score only through a training-fold-selected guarded residual.

# 4. NB12b confirmation and frozen model

NB12b fixed one eligible challenger, one protected baseline, three fresh seeds, the fusion grid, and 12 acceptance conditions before evaluation. Seed 271 remained discovery evidence; seeds 272, 273, and 274 formed the confirmation.

## Fresh-seed stability

| Seed | Compact PR-AUC | Guarded PR-AUC | Gain | Improved countries |
|---|---|---|---|---|
| 272 | 0.9065 | 0.9300 | +0.0234 | 5 of 6 |
| 273 | 0.9056 | 0.9284 | +0.0228 | 4 of 6 |
| 274 | 0.9086 | 0.9304 | +0.0218 | 6 of 6 |

## Operational review budget

At a 20% review budget, the compact baseline recovered 55 of 87 known positives across the six country panels. Guarded fusion recovered 54. Libya lost one known positive; all other countries were unchanged. This met the fixed maximum loss of one source per country.

## Frozen artifact contract

- Nine compact LightGBM models: three folds for each of seeds 272, 273, and 274.
- Three visual positive-unlabelled models and one fusion calibration archive.
- Fusion alpha 0.5 for each full-foreign seed pipeline.
- Hashed CV embeddings, morphology features, schemas, thresholds, and predictions.
- All 33 artifacts listed in the manifest passed hash verification.

## Decision

Freeze `guarded_cv_tabular` as the selected foreign-country ensemble. Keep the compact tabular branch as the protected fallback when imagery is missing or an integrity check fails. The all-77 branch remains diagnostic only.

## Runtime note

The CPU confirmation completed in about 3.14 minutes because it reused audited embedding and morphology caches. That time is not an end-to-end image-encoding benchmark.

# 5. India status, limitations, and next work

## Historical India result

| Model | Sources | Precision | Recall | F1 | PR-AUC | ROC-AUC |
|---|---|---|---|---|---|---|
| Superseded NB8 | 706,686 | 0.093 | 0.391 | 0.151 | 0.142 | 0.915 |

The historical run contained 197 EOG-matched source rows. It detected 49 of 97 recoverable EOG sites, equal to 50.5% recall of recoverable sites and 26.2% recall of all 187 active sites used in that run.

The final NB12b guarded model has not been evaluated on India. A future run must use the frozen model without retuning and must be described as a final-model transfer evaluation, not as a first untouched test.

## Limitations

- EOG labels cover gas flares, not every industrial thermal-source category.
- The 294-source multimodal panel is EOG-enriched and does not estimate population precision.
- Three fresh seeds measure algorithmic variation, not new-country sampling variation.
- The bootstrap interval crosses zero, so the positive visual gain remains uncertain.
- Candidate construction limits site recall, especially in Angola and Indonesia.
- Facility absence cannot establish that an unmatched high-score source is false.

## Next work

- Run frozen NB12b transfer inference on India with no threshold or model changes.
- Create a population-scale review queue and estimate precision by spatially stratified manual review.
- Separate flare, kiln, cement, steel, crop-burning, and wildfire error modes.
- Improve upstream candidate coverage before claiming facility-level recall.

## Conclusion

The defensible result is a guarded hybrid that improves ranking on the fixed foreign panel while preserving a compact tabular fallback. The project has a reproducible frozen artifact set and an honest account of what remains unproven.